

Next Steps — GCR Dosimetry Follow-On

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1 9/25/26 — where things stand

`gcrisk` (formerly `gcr-dosimetry-pipeline`) is submitted to *Life Sciences in Space Research*. Manuscript number assigned, build-for-approval PDF came back clean — no compile log dump, all 7 figures rendering with real data, all cross-refs/citations resolved. Also found and fixed a real repo problem today: local `main` was 11 commits ahead of GitHub and 2 behind (a stale README edit made directly on GitHub back on 6/4/26, before the `gcrisk` rename and the ISS/RAD validation work). Merged, caught the merge silently reverting one sentence to the old typo'd version, fixed it, pushed. Repo is clean and in sync now, including the full EM submission package (`paper/submission/`) so the exact files submitted are preserved in version control.

So: nothing blocking, no open repo issues. This document is just about what's worth doing next, not anything urgent.

2 9/25/26 — skimmed two ScienceDirect exports for next-project ideas

Two unrelated clusters, both from the same *LSSR* special issue:

1. **Astrochemistry / prebiotic ice chemistry** (13 papers) — COMs in protostellar systems, nucleobases in interstellar ices, glycine stability under ionizing radiation, cosmic-ray implantation into molecular clouds. Interesting field, but a genuinely different research track from anything `gcrisk` does. Not pursuing unless I deliberately want to branch into astrobiology/astrochemistry, which is a separate decision from “what's the sequel to the GCR paper.”

2. **Space radiation dose-response / quality-factor modeling** (14 papers) — this is the one that matters. Several of these are direct hits against the exact uncertainty structure the just-submitted paper already quantified.

The overlap that matters: the paper’s own value-of-information sweep found that halving Q_{factor} uncertainty alone shrinks p95 REID by 33.5% — about $7\times$ more leverage than any other single parameter, more than every GCR transport-physics parameter combined. That’s not a throwaway finding, it’s a pointer at exactly what to work on next, and this paper batch landed with several candidate answers to it.

3 Candidate next projects, ranked

Idea	What it is	Effort
Charge-weighted Q^*	Naito & Kodaira propose Q^* , a GCR-specific quality factor based on LET and charge that diverges from ICRP-60 $Q(L)$ at high LET, claiming current dose assessments over-weight the heavy-ion contribution. Direct drop-in alternative to what <code>reid.py</code> already uses.	Low
Ensemble dose-response model	Slaba & Poignant (NASA Langley) — a probabilistic, non-linear QF framework meant to replace the Cucinotta-era calibration approach the pipeline currently inherits. This is the actual research-frontier answer, not a swap-in.	High
ICRP $Q(L)$ history/meta-analysis	Sato’s review — not a project, but a citation that directly backs why Q -factor is the dominant uncertainty source. Should already be in the lit review.	Trivial
Murine organ DCFs + gastric cancer	Hosseini et al. (organ dose conversion factors for GCRsim mice) and Suman et al. (dose/LET-dependent gastric tumor incidence in Apc1638N/+ mice) — real GCRsim-derived organ-dose and tumor-incidence data. Interesting as a validation target if REID validation ever moves past human epidemiological life tables into direct dose-to-tumor-incidence data.	Medium, bigger scope

Pick: Q^* . Best ratio of effort to payoff — it’s a direct, quantifiable extension of the exact sensitivity result the submitted paper already highlights, reuses the existing

LHS/REID machinery almost unchanged, and gives a natural “part 2” paper: same validated pipeline, one swapped component, a concrete number for how much REID moves.

4 Build on top of gcrisk, or start separate?

On top of gcrisk. The whole value of the Q^* idea is that it’s the same validated dose/REID/LHS pipeline with one component swapped — forking a new repo means re-deriving the transport and validation harness for no benefit.

Risk: bolting a second quality-factor model directly into `reid.py` muddies the story of a package the just-submitted paper already describes in print. Mitigation: keep it in a clearly separate module rather than a rewrite in place —

- `gcrisk/qf_models.py` — `qf_icrp60(LET)` (existing logic, moved/kept as-is) and `qf_naito_kodaira(LET, Z)` (new).
- `reid.py` takes a `qf_model` argument rather than hard-coding ICRP-60.
- Version bump/tag so the new paper can just say “`gcrisk vX.Y`, run with QF model = Q^* ” — core package stays legible, second paper stays citable and reproducible off a pinned version.

5 Concrete next action

Not started yet — this is the actual to-do, next time I sit down with this:

1. Pull the exact $Q^*(LET, Z)$ formula and Z -contribution weighting out of the Naito & Kodaira PDF (*Charge-weighted quality factors based on LET for practical dose assessment of space radiation*).
2. Implement `qf_naito_kodaira()` in a new `gcrisk/qf_models.py`, unit-tested against the paper’s own reported Q^* values at a few LET points before touching the ensemble.
3. Add the `qf_model` switch to `reid.py` / `uncertainty.py` without changing default behavior (ICRP-60 stays default).
4. Re-run the N=500 LHS ensemble with Q^* in place of ICRP-60 at the same 16 g/cm² Al calibration geometry already used in the submitted paper.
5. Report the shift: median REID, p5–p95 REID, and how much of the 75% Q_{factor} -driven variance share moves. That number is the whole paper.